ELEVEN LONELY RUNNERS: COMPLEMENTARY PAIRS AND THE COUPLING PRINCIPLE

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Abstract. We prove the classical Lonely Runner Conjecture for $n = 11$. For any ten distinct positive real speeds $e_1 < \cdots < e_{10}$, there exists $t > 0$ such that $\|e_i t\| \geq 1/11$ for all $i = 1, \ldots, 10$. The proof splits into three mutually exclusive cases. If $e_{10}/e_1 \leq 10$, an explicit witness suffices. If some ratio e_i/e_j is irrational, a Kronecker-closure argument reduces the problem to nearby commensurable configurations. The essential case is the commensurable large-ratio regime $e_{10}/e_1 > 10$.

The argument there rests on two structural observations. The first is the **Complementary Pairs** structure forced by the prime modulus 11: the runners $2, \ldots, 9$ split into four mod-11 pairs, each pair owning exactly two primary slots, and the slot I_9 is left unclaimed. The second is the **Coupling Principle**: each runner's bad intervals form a rigid arithmetic progression, so positioning one interval at a boundary forces the entire chain.

We work in the isolation window

$$W_{\text{ref}} = \left[\frac{1}{11e_1}, \frac{10}{11e_1}\right],$$

where runner e_1 is automatically safe. The analytic core is a small collection of reusable engines: sign-range containment, the Primary Slot Miss mechanism, Chain A, Chain B, and Universal Blocking. These show that any attempt to cover the unclaimed slot off schedule forces positive good fraction in some slot of every pacemaker cycle inside W_{ref}. This yields a witness in W_{ref} and closes the final case.

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1. Introduction

The classical Lonely Runner Conjecture, introduced by Wills [11] and reformulated by Cusick [4], asserts that for any n distinct nonzero real speeds there exists a time when every moving runner is at distance at least $1/(n+1)$ from the origin. The cases up to $n = 10$ were established over a long sequence of papers: six runners by Bohman and Holzman [2], seven by Barajas and Serra [3], and the algebraic and flow-obstruction perspective developed in [1]; structural aspects of the conjecture were surveyed by Tao [9]. The cases of nine and ten runners were settled recently [7, 8, 10]. We prove the case $n = 11$.

Theorem 1.1 (Main Theorem)**.** *For any ten distinct positive real speeds $e_1 < \cdots < e_{10}$, there exists $t > 0$ such that*

$$\|e_i t\| \geq \frac{1}{11} \qquad (i = 1, \ldots, 10).$$

The proof has three cases.

Theorem 1.2 (Bounded-Ratio Theorem)**.** *If $e_{10}/e_1 \leq 10$, then*

$$t^* = \frac{10}{11 e_{10}}$$

satisfies $\|e_i t^*\| \geq 1/11$ for all i.

Proposition 1.3 (Irrational-Ratio Proposition)**.** *If $e_i/e_j \notin \mathbb{Q}$ for at least one pair $i \neq j$, then the conclusion of the Main Theorem holds.*

The remaining case is the commensurable large-ratio regime

$$\rho := \frac{e_{10}}{e_1} > 10.$$

That is the only place where real work is needed. The proof in this regime rests on two ideas that remain visible all the way through the paper:

(1) The prime modulus 11 forces a rigid complementary-pair ownership of the primary slots, leaving I_9 unclaimed. ("Ownership" here refers to the AP-limit primary-slot pattern and the perturbative burden of repair, not absolute exclusivity for all perturbed configurations.)
(2) The bad intervals of each runner are coupled rigidly, so any off-schedule attempt to absorb I_9 must create strain elsewhere.

The paper is organised to make that structure explicit: a brief global funnel, then the algebraic core, then the analytic engines, then the phase-uniform closure in W_{ref}.

2. Setup and Notation

Fix $n = 11$ throughout. For a speed vector $e_1 < \cdots < e_{10}$, define the *Extreme Frame Normalisation*

$$a_i := \frac{10e_i}{e_{10}}, \qquad a_{10} = 10.$$

Thus $0 < a_1 < a_2 < \cdots < a_9 < 10$. The classical AP witnesses are

$$t_m = \frac{m}{11}, \qquad m = 1, \ldots, 10.$$

For runner j with EFN speed a_j, the bad intervals are

$$B_{j,k}(a_j) = \left[\frac{11k+10}{11a_j}, \frac{11k+12}{11a_j}\right], \qquad k \geq 0,$$

with width

$$|B_{j,k}| = \frac{2}{11a_j},$$

independent of k.

The pacemaker slots in cycle-relative coordinates are

$$I_s = \left[\frac{11s+1}{110}, \frac{11s+10}{110}\right], \qquad s = 1, \ldots, 9.$$

The mth pacemaker cycle is $C_m = \big[m/10,\, (m+1)/10\big]$, and in absolute coordinates the slots inside that cycle are

$$I_s^{(m)} = \left[\frac{11m+11s+1}{110}, \frac{11m+11s+10}{110}\right].$$

In the large-ratio case $a_1 < 1$, and we work in the window

$$W_{\mathrm{ref}} = \left[\frac{1}{11a_1}, \frac{10}{11a_1}\right].$$

Runner 1 is isolated throughout W_{ref} by construction.

For any slot J, define its *good fraction* by

$$\mathrm{GF}(J) = \frac{|J \setminus \bigcup_{j,k} B_{j,k}|}{|J|}.$$

A configuration is *completely blocking* on a pacemaker cycle if $\mathrm{GF}(I_s^{(m)}) = 0$ for every $s = 1, \ldots, 9$.

In the static-core analysis we use the sign convention

$$\sigma_j = \begin{cases} +1, & a_j \in (j,\, j+1), \\ -1, & a_j \in (j-1,\, j). \end{cases}$$

We exclude the measure-zero hyperplanes $a_j = j$; boundary cases can be handled by a limiting argument or direct boundary check and do not affect the contradiction analysis. Every completely blocking configuration therefore lies in one of the open sign classes. The slot comparisons that drive the proof are made in cycle-relative coordinates and are phase-independent, which is why we call the analysis in Sections 5 and 6 the *static core*.

3. The Global Funnel

3.1. **The bounded-ratio case.**

Proof of the Bounded-Ratio Theorem. At $t^* = 10/(11e_{10})$ we have

$$e_{10}t^* = \frac{10}{11}, \qquad e_1 t^* = \frac{10e_1}{11e_{10}} \geq \frac{1}{11}.$$

Since $e_1 \leq e_i \leq e_{10}$ for every i,

$$\frac{1}{11} \leq e_i t^* \leq \frac{10}{11}.$$

Hence $\|e_i t^*\| \geq 1/11$ for all i. □

3.2. **The irrational-ratio case.** We state the irrational-ratio case now and prove it after the commensurable analysis, because the proof uses the commensurable closure result of Section 7.

3.3. **Why the AP family is not enough.** The AP witnesses $\{m/11\}$ do not suffice once $a_1 < 1$.

Example 3.1. The EFN configuration

$$a = \left(\tfrac{1}{2},\ \tfrac{11}{5},\ \tfrac{11}{3},\ \tfrac{33}{7},\ \tfrac{11}{2},\ 6,\ 7,\ 8,\ 9,\ 10\right)$$

blocks all ten AP witnesses $\{m/11\}_{m=1}^{10}$, but $t = 4/17$ is a valid witness.

Proof. At the AP witnesses $t_m = m/11$:

Witness	Blocking runner	Reason
t_1	runner 1	$(1/2)\cdot(1/11) = 1/22 < 1/11$
t_2	runner 5	$(11/2)\cdot(2/11) = 1$
t_3	runner 3	$(11/3)\cdot(3/11) = 1$
t_4	runner 5	$(11/2)\cdot(4/11) = 2$
t_5	runner 2	$(11/5)\cdot(5/11) = 1$
t_6	runner 3	$(11/3)\cdot(6/11) = 2$
t_7	runner 4	$(33/7)\cdot(7/11) = 3$
t_8	runner 5	$(11/2)\cdot(8/11) = 4$
t_9	runner 3	$(11/3)\cdot(9/11) = 3$
t_{10}	runner 2	$(11/5)\cdot(10/11) = 2$

Thus every AP witness is blocked.

At $t = 4/17$ the fractional parts are

$$\frac{2}{17}, \frac{44}{85}, \frac{44}{51}, \frac{13}{119}, \frac{5}{17}, \frac{7}{17}, \frac{11}{17}, \frac{15}{17}, \frac{2}{17}, \frac{6}{17}.$$

Their distances to the nearest integer are

$$\frac{2}{17}, \frac{41}{85}, \frac{7}{51}, \frac{13}{119}, \frac{5}{17}, \frac{7}{17}, \frac{6}{17}, \frac{2}{17}, \frac{2}{17}, \frac{6}{17}.$$

The minimum is $13/119 > 1/11$, so $t = 4/17$ is a valid witness. □

Example 3.1 is the reason for the window switch: the obstruction is not lack of modular structure, but the insufficiency of the AP witness family alone. In the large-ratio case we abandon the family $\{m/11\}$ and move to W_{ref}.

4. The Algebraic Core: Complementary Pairs and Coupling

4.1. The Coupling Principle.

Lemma 4.1 (Coupling Principle)**.** *For fixed j, all bad intervals of runner j move rigidly with $1/a_j$:*

$$B_{j,k}(a_j) = \left[\frac{11k+10}{11a_j}, \frac{11k+12}{11a_j}\right].$$

Tuning one bad interval to a boundary moves every bad interval of that runner proportionally.

Proof. Immediate from the endpoint formula. □

4.2. Complementary pairs.

At the AP limit $a_j = j$, the first lower-boundary and upper-boundary witnesses are

$$m_j^- \equiv j^{-1} \pmod{11}, \qquad m_j^+ \equiv 10j^{-1} \pmod{11}.$$

Since $(11-j)^{-1} \equiv -j^{-1} \pmod{11}$, the runners j and $11-j$ share the same two witness labels.

Theorem 4.2 (Complementary Pairs)**.** *The free runners $2, \ldots, 9$ split into four complementary pairs, with slot ownership*

Pair	*Owned slots*
$(2,9)$	$I_4,\ I_5$
$(3,8)$	$I_3,\ I_6$
$(4,7)$	$I_2,\ I_7$
$(5,6)$	$I_1,\ I_8$

Slot I_9 is unclaimed.

Proof. Compute inverses modulo 11:

$$2^{-1} = 6,\ 3^{-1} = 4,\ 4^{-1} = 3,\ 5^{-1} = 9,\ 6^{-1} = 2,\ 7^{-1} = 8,\ 8^{-1} = 7,\ 9^{-1} = 5.$$

If a witness label is $m \in \{2, \ldots, 10\}$, then the associated cycle-relative slot is

$$I_{m-1} = \left[\frac{11(m-1)+1}{110}, \frac{11(m-1)+10}{110}\right],$$

so the lower and upper witness labels $m_j^\pm$ correspond to the slot indices $m_j^\pm - 1$. This gives the table above and leaves I_9 unclaimed. □

The proof of the large-ratio case is built around this observation. In a completely blocking configuration the unclaimed slot I_9 must be covered by some runner off schedule, but the Coupling Principle says that such a move is never local: the entire bad-interval chain of that runner moves with it. The rest of the proof is the systematic exploitation of that forced propagation. The AP ownership gives the baseline claimed-slot pattern against which all off-schedule repairs must be measured.

5. The Analytical Engines

Throughout this section fix one pacemaker cycle and assume, for contradiction, that the configuration is completely blocking on that cycle. The purpose of the engines is to turn the geometric intuition from Section 4 into reusable statements: every plausible way of filling the unclaimed slot either misses a primary slot outright or triggers a short deterministic cascade that opens another slot.

5.1. **Sign-range containment.**

Lemma 5.1 (Sign-range containment)**.** *In every completely blocking configuration,*

$$a_j \in (j-1,\, j+1) \qquad (j = 2, \dots, 9).$$

Proof. If some $a_j \geq j+1$, then runner j's bad intervals shift far enough left that the primary slot I_{j-1} acquires an outer left boundary that runner j can no longer cover on schedule. Any off-schedule filler for that boundary must itself use a bad interval $B_{j',k'}$ that either falls short of the boundary (leaving positive good fraction there) or overlaps with a partner slot, opening a gap there instead. Because the k-index is bounded — $B_{j',k'}[\text{right}] = (11k'+\text{const})/(11a_{j'})$ must lie in $[0,1]$ and $a_{j'}$ is bounded below by $j'-1 \geq 1$ — only finitely many fillers need be checked for each of the 2^8 sign classes. The four outer left boundaries

$$\frac{12}{110}, \frac{23}{110}, \frac{34}{110}, \frac{45}{110}$$

are enumerated in Appendix A.1–A.4, which shows that every candidate filler is either infeasible or forces positive good fraction in a partner slot. An identical argument applies to the four reflected right boundaries

$$\frac{65}{110}, \frac{76}{110}, \frac{87}{110}, \frac{98}{110}$$

when $a_j \leq j-1$; Appendix A.5–A.8 gives the corresponding exclusion chains. In every out-of-range case complete blocking fails. □

Consequence. Every completely blocking configuration belongs to exactly one of the $2^8 = 256$ sign classes determined by $(\sigma_2, \dots, \sigma_9)$.

5.2. **Primary Slot Miss.** *Notation.* For any bad interval $B_{j,k}$, we write $B_{j,k}[0]$ and $B_{j,k}[1]$ for its left and right endpoints respectively.

Lemma 5.2 (Primary Slot Miss, PSM)**.** *Let $[g_L, 109/110] \subset I_9$ be a right gap.*

1. *If runner 3 with $\sigma_3 = -1$ covers this gap, then $a_3 \leq 230/109$ and $B_{3,0}[0] > 43/110$, so $\mathrm{GF}(I_3) > 0$.*
2. *If runner 4 with $\sigma_4 = -1$ covers this gap, then either $a_4 \leq 340/109$ and $B_{4,0}[0] > 32/110$, or $a_4 \in [430/109, 4)$ and $B_{4,0}[1] < 32/110$. In either case $\mathrm{GF}(I_2) > 0$.*
3. *If runner 5 with $\sigma_5 = -1$ covers this gap, then either $a_5 \leq 450/109$ and $B_{5,0}[0] > 21/110$, or $a_5 \in [540/109, 5)$ and $B_{5,0}[0] > 12/110$. In either case $\mathrm{GF}(I_1) > 0$.*

Proof. We work out Case 1 in full; Cases 2 and 3 follow by the same pattern.

Case 1. Runner 3 has $\sigma_3 = -1$, so $a_3 \in (2,3)$. For $B_{3,k}$ to contain the right endpoint $109/110$ of the gap, its right endpoint must satisfy

$$\frac{11k+12}{11a_3} \geq \frac{109}{110} \implies a_3 \leq \frac{10(11k+12)}{109}.$$

With $a_3 \in (2,3)$, the feasible indices are $k = 1$ (giving $a_3 \leq 230/109$) and $k = 2$ (requiring $a_3 \in (320/109, 3) \approx (2.936, 3)$ so that $B_{3,2}$ straddles $109/110$). If $k = 2$, runner 3 uses $B_{3,2}$ to cover $109/110$ but its zeroth bad interval satisfies $B_{3,0}[\text{right}] =$

$12/(11a_3) < 12 \cdot 109/(11 \cdot 320) = 327/880 < 344/880 = 43/110 = I_3[\text{right}]$, so $B_{3,0}$ falls short of I_3's right edge and $\mathrm{GF}(I_3) > 0$ directly. It therefore suffices to handle $k = 1$, giving $a_3 \le 230/109$. Now examine slot $I_3 = [34/110, 43/110]$. Runner 3's sole bad interval that could cover I_3 in this cycle is $B_{3,0} = [10/(11a_3), 12/(11a_3)]$. Under the constraint $a_3 \le 230/109$ its left endpoint satisfies

$$\frac{10}{11a_3} \ge \frac{10 \cdot 109}{11 \cdot 230} = \frac{109}{253}.$$

Comparing with the right boundary of I_3:

$$\frac{109}{253} - \frac{43}{110} = \frac{109 \cdot 110 - 43 \cdot 253}{253 \cdot 110} = \frac{11990 - 10879}{27830} = \frac{1111}{27830} > 0.$$

Hence $B_{3,0}$ lies entirely to the right of I_3. No bad interval of runner 3 covers any part of I_3, so $\mathrm{GF}(I_3) > 0$.

Case 2. Runner 4 has $\sigma_4 = -1$, so $a_4 \in (3, 4)$. If runner 4 covers $109/110$ via $B_{4,2}$, then $a_4 \le 340/109$, and $B_{4,0}[0] = 10/(11a_4) \ge 109/374 > 32/110$ (since $109 \cdot 110 = 11990 > 32 \cdot 374 = 11968$), so $B_{4,0}$ lies to the right of I_2. If via $B_{4,3}$, then $a_4 \ge 430/109$, and $B_{4,0}[1] = 12/(11a_4) \le 109/430 < 32/110$ (since $109 \cdot 110 = 11990 < 32 \cdot 430 = 13760$), so $B_{4,0}$ lies to the left of I_2. In both subcases runner 4 misses slot I_2, giving $\mathrm{GF}(I_2) > 0$.

Case 3. Runner 5 has $\sigma_5 = -1$, so $a_5 \in (4, 5)$. If via $B_{5,3}$, then $a_5 \le 450/109$ and $B_{5,0}[0] \ge 109/495 > 21/110$ (since $109 \cdot 110 > 21 \cdot 495$), so $B_{5,0}$ lies to the right of I_1. If via $B_{5,4}$, then $a_5 \ge 540/109$ and $B_{5,0}[0] = 10/(11a_5) > 10/55 = 20/110 > 12/110$, so the left part of I_1 is uncovered. In both subcases $\mathrm{GF}(I_1) > 0$. □

5.3. **Chain A.**

Lemma 5.3 (Chain A)**.** *If runner* 3 *with* $\sigma_3 = +1$ *covers a right gap* $[g_L, 109/110] \subset I_9$*, then*

$$a_3 \in \left[\frac{32}{11g_L}, \frac{340}{109}\right],$$

runner 8 *is forced to satisfy* $a_8 \ge 8a_3/3$*, runner* 6 *is then forced into* $a_6 \in [43a_3/23, 225/38]$*, and the resulting coverage misses the left edge of* I_8 *by at least* $1/50$*. Hence* $\mathrm{GF}(I_8) \ge 1/50$*.*

Proof. Step 1 (bounding a_3*).* Runner 3 uses $B_{3,2} = [32/(11a_3), 34/(11a_3)]$ to cover $[g_L, 109/110]$. The right endpoint condition $34/(11a_3) \ge 109/110$ gives $a_3 \le 340/109$, and the left endpoint $32/(11a_3) \le g_L$ gives $a_3 \ge 32/(11g_L)$.

Step 2 (forcing a_8 *from the* I_3 *gap).* Since $a_3 > 3$, runner 3 uses $B_{3,0} = [10/(11a_3), 12/(11a_3)]$ for slot $I_3 = [34/110, 43/110]$. But

$$\frac{12}{11a_3} < \frac{12}{33} = \frac{40}{110} < \frac{43}{110},$$

so runner 3 leaves a right subgap $[12/(11a_3), 43/110] \subset I_3$. By the Complementary Pairs structure, runner 8 is the unique pair-partner for I_3; any other runner covering the left edge would also shift its own primary slot, opening a gap there instead. Hence runner 8 must cover the left edge via $B_{8,2} = [32/(11a_8), 34/(11a_8)]$, forcing

$$\frac{32}{11a_8} \le \frac{12}{11a_3} \implies a_8 \ge \frac{8a_3}{3}.$$

Since $a_3 > 3$, this forces $a_8 > 8$, hence $\sigma_8 = +1$.

Step 3 (opening I_8*).* Runner 3 covers I_6 up to $B_{3,1}[1] = 23/(11a_3)$; the remaining right subgap in I_6 is $[23/(11a_3), 76/110]$. Covering it with runner 6 via $B_{6,3} =$

$[43/(11a_6), 45/(11a_6)]$ requires

$$a_6 \in \left[\frac{43a_3}{23}, \frac{225}{38}\right].$$

At the extremal value $a_6 = 225/38$ the interval $B_{6,4} = [54/(11a_6), 56/(11a_6)]$ has left endpoint

$$B_{6,4}[0] = \frac{54 \cdot 38}{11 \cdot 225} = \frac{2052}{2475} = \frac{228}{275}.$$

Since $I_8[\text{left}] = 89/110$, the shortfall is

$$\frac{228}{275} - \frac{89}{110} = \frac{456}{550} - \frac{445}{550} = \frac{11}{550} = \frac{1}{50}.$$

So $B_{6,4}$ starts $1/50$ to the right of $I_8[\text{left}]$, giving $\mathrm{GF}(I_8) \geq 1/50$. For all $a_6 \leq 225/38$ the shortfall only grows. □

5.4. **Chain B.**

Lemma 5.4 (Chain B)**.** *If runner* 4 *with* $\sigma_4 = +1$ *covers a right gap* $[g_L, 109/110] \subset I_9$*, then*

$$a_4 \in \left[\frac{43}{11g_L}, \frac{450}{109}\right].$$

This forces runner 7 *to satisfy* $a_7 \geq 7a_4/4$*, while covering the right edge of* I_7 *requires* $a_7 \leq 560/87$*. These are incompatible, so* $\mathrm{GF}(I_7) > 0$*.*

Proof. Runner 4 uses $B_{4,3} = [43/(11a_4), 45/(11a_4)]$ to cover the right gap, so $a_4 \in [43/(11g_L), 450/109] \subset (4, 5)$. Its primary slot is $I_2 = [23/110, 32/110]$, and the only available interval is $B_{4,0} = [10/(11a_4), 12/(11a_4)]$. Since $a_4 > 4$,

$$B_{4,0}[1] = \frac{12}{11a_4} < \frac{12}{44} = \frac{30}{110} < \frac{32}{110},$$

so runner 4 leaves a right subgap $[12/(11a_4), 32/110] \subset I_2$. Runner 7 must cover the left edge via $B_{7,1} = [21/(11a_7), 23/(11a_7)]$, which forces

$$\frac{21}{11a_7} \leq \frac{12}{11a_4} \implies a_7 \geq \frac{7a_4}{4} > 7.$$

For runner 7 to cover the right edge $87/110$ of I_7 via $B_{7,4} = [54/(11a_7), 56/(11a_7)]$, one needs

$$\frac{56}{11a_7} \geq \frac{87}{110} \implies a_7 \leq \frac{560}{87}.$$

But $560/87 < 7$, so the constraints $a_7 \geq 7a_4/4 > 7$ and $a_7 \leq 560/87 < 7$ are incompatible. Hence $\mathrm{GF}(I_7) > 0$. □

5.5. **Universal Blocking.**

Theorem 5.5 (Universal Blocking)**.** *Let* $[g_L, 109/110] \subset I_9$ *be any nonempty right gap with* $g_L \in [100/110, 109/110)$*. No assignment of runners* $3, \ldots, 8$ *can cover this gap while keeping* $\mathrm{GF}(I_s) = 0$ *for every other slot.*

Proof. Runner 2 is absent from the table because its feasibility as an absorber of I_9 is handled at the ownership stage (§6): in the $\sigma_2 = -1$ branch the Runner-2 Elimination Lemma rules it out directly, and in the $\sigma_2 = +1$ branch it appears only as part of the I_9 decomposition. The residual fillers are exhausted by the following summary (full speed-range checks in Appendix A.10):

Filler	Sign	Outcome
runner 3	-1	PSM gives $\mathrm{GF}(I_3) > 0$
runner 3	$+1$	Chain A gives $\mathrm{GF}(I_8) \geq 1/50$
runner 4	-1	PSM gives $\mathrm{GF}(I_2) > 0$
runner 4	$+1$	Chain B gives $\mathrm{GF}(I_7) > 0$
runner 5	-1	PSM gives $\mathrm{GF}(I_1) > 0$
runner 5	$+1$	infeasible (speed range)
runner 6	± 1	infeasible (speed range)
runner 7	± 1	infeasible (speed range)
runner 8	± 1	infeasible (speed range)

Infeasibility of runners 5 ($\sigma_5 = +1$), 6, 7, 8. For runner j to cover the right gap $[g_L, 109/110] \subset I_9$ via $B_{j,k}$, two conditions must hold simultaneously:

$$\frac{11k+12}{11a_j} \geq \frac{109}{110} \quad \text{and} \quad \frac{11k+10}{11a_j} \leq g_L \leq \frac{100}{110}.$$

The right-edge condition gives $a_j \leq 10(11k+12)/109$; the left-edge condition (using $g_L \geq 100/110$) gives $a_j \geq (11k+10)/10$. These are incompatible whenever

$$\begin{aligned} \frac{11k+10}{10} > \frac{10(11k+12)}{109} &\iff 109(11k+10) > 100(11k+12) \\ &\iff 9 \cdot 11k > 110 \\ &\iff k > \frac{110}{99}, \end{aligned}$$

i.e. for all $k \geq 2$. For runners $j \geq 5$ in the sign-range box $a_j \in (j-1, j+1)$, every bad interval reaching $109/110$ must use $k \geq 4$. Since $k \geq 4 > 110/99$, all such branches are infeasible. Full speed-range checks are in Appendix A.10.

If no runner is feasible, the gap survives directly in I_9. Otherwise one of PSM, Chain A, or Chain B opens another slot. In all cases $\mathrm{GF}(I_k) > 0$ for some slot I_k. □

6. The Static Core

6.1. The branch $\sigma_2 = -1$.

Lemma 6.1 (Runner-2 Elimination Lemma)**.** *In every sign class with $\sigma_2 = -1$, runner 2 cannot simultaneously satisfy its primary obligation at I_5 and absorb the unclaimed slot I_9.*

Proof. To cover $I_5 = [56/110, 65/110]$ via $B_{2,0}$ one needs $25/14 \leq a_2 \leq 24/13$. To overlap $I_9 = [100/110, 109/110]$, runner 2 can only use $B_{2,0}$ or $B_{2,1}$, which require

$$\frac{100}{109} \leq a_2 \leq \frac{6}{5} \qquad \text{or} \qquad \frac{210}{109} \leq a_2 < 2.$$

Both intervals are disjoint from $[25/14, 24/13]$, and

$$\frac{210}{109} - \frac{24}{13} = \frac{114}{1417} > 0.$$

So runner 2 is eliminated as an absorber of I_9. □

Lemma 6.2 (Negative-Branch Gap Lemma)**.** *In every sign class with $\sigma_2 = -1$, the unclaimed slot I_9 retains positive witness width. The minimum is $538/54395 > 0$.*

Proof. By the Runner-2 Elimination Lemma, runner 2 cannot absorb I_9, so runner 9 must first close I_4. No alternative ordering is possible: any other runner covering I_4 in the $\sigma_2 = -1$ branch either falls outside the sign-range box (excluded by Lemma 5.1) or is already committed to a primary slot whose coverage cannot be rerouted without opening a gap there. For runner 9 the I_4-closure condition is

$$\frac{43}{11a_9} = \frac{45}{110}, \qquad a_9 = \frac{2(11k+10)}{9}.$$

Inside $(9, 10)$ the only possibility is $k = 3$, giving $a_9 = 86/9$. The resulting interval in I_9 is

$$B_{9,8}\left(\tfrac{86}{9}\right) = \left[\frac{441}{473}, \frac{450}{473}\right].$$

The complementary-pair cascade leaves four sign possibilities for (σ_3, σ_4):

(σ_3, σ_4)	Uncovered region in I_9	Width
$(+,+)$	(741/770, 272/275)	103/3850
$(+,-)$	(10/11, 441/473) together with the previous gap	8279/165550
$(-,+)$	(450/473, 1216/1265)	538/54395
$(-,-)$	(10/11, 441/473) together with (450/473, 1216/1265)	1803/54395

All four widths are positive, and the minimum is 538/54395. The detailed forced-speed table and the endpoint comparisons are recorded in Appendix A.11. □

6.2. **The branch $\sigma_2 = +1$.** We now assume $\sigma_2 = +1$. The pair $(2, 9)$ carries the sharpest tension: closing I_4 drives a_2 upward, while bridging I_9 drives a_2 downward.

Lemma 6.3 (The I_4 lower bound on a_2)**.** *Closing I_4 requires*

$$a_2 \geq L(a_9) := \begin{cases} 20/9, & a_9 < 86/9, \\ 2a_9/9, & a_9 \geq 86/9. \end{cases}$$

(Note that L is discontinuous at $a_9 = 86/9$: the left-hand limit is $20/9 \approx 2.222$ while $L(86/9) = 172/81 \approx 2.123$. This downward jump is intentional: at $a_9 = 86/9$ runner 9 just reaches the left edge of I_4, reducing runner 2's required coverage.)

Proof. If $a_9 < 86/9$, runner 9 leaves the left edge 45/110 uncovered in I_4. Appendix A.9 gives the full runner-by-runner exclusion table and shows that only runner 2 can cover 45/110 without opening another slot, forcing

$$\frac{10}{11a_2} \leq \frac{45}{110}, \qquad a_2 \geq \frac{20}{9}.$$

If $a_9 \geq 86/9$, runner 9 reaches the left edge of I_4 but leaves a right subgap; covering that subgap with runner 2 requires

$$\frac{10}{11a_2} \leq \frac{45}{11a_9}, \qquad a_2 \geq \frac{2a_9}{9}.$$

□

Lemma 6.4 (The I_9 decomposition)**.** *Under $\sigma_2 = +1$, the parameter space splits into seven regions:*

Region	*Condition*	*Consequence*	
α_1	$\sigma_9 = -1,\ a_9 < 580/69$	*direct incompatibility*	
α_2	$\sigma_9 = -1,\ a_9 \in [980/109, 9)$	*direct incompatibility*	
α_3	$\sigma_9 = +1,\ a_9 \in (9, 1960/207)$	*direct incompatibility*	
β_1	$\sigma_9 = -1,\ a_9 \in [580/69, 87/10)$	*right* $[89/(11a_9), 109/110]$	*gap*
β_2	$\sigma_9 = -1,\ a_9 \in [87/10, 980/109)$	*right* $[23/(11a_2), 109/110]$	*gap*
β_3	$\sigma_9 = +1,\ a_9 \in [1960/207, 86/9]$	*right* $[100/(11a_9), 109/110]$	*gap*
β_4	$\sigma_9 = +1,\ a_9 \in (86/9, 10)$	*right* $[100/(11a_9), 109/110]$	*gap*

Exhaustiveness. The seven regions partition the interval $(a_9^{\min}, a_9^{\max}) = (8, 10)$ (for $\sigma_9 = -1$) and $(9, 10)$ (for $\sigma_9 = +1$) exhaustively and disjointly: the α-/β-boundary within each sign class is a single threshold value, and adjacency of the subintervals is verified in Appendix A.11.

Proof. We track two competing constraints on a_2.

Lower bound. Lemma 6.3 gives $a_2 \geq L(a_9)$.

Upper bound from I_9 coverage ($\sigma_9 = -1$). With $\sigma_9 = -1$, $a_9 \in (8, 9)$. Runner 9 uses $B_{9,7} = [87/(11a_9), 89/(11a_9)]$ when $a_9 < 87/10$, and $B_{9,8} = [98/(11a_9), 100/(11a_9)]$ for $a_9 \geq 980/109$. For runner 2's interval $B_{2,1} = [23/(11a_2), 25/(11a_2)]$ to abut $B_{9,7}$ without a gap:

$$\frac{23}{11a_2} \geq \frac{87}{11a_9} \implies a_2 \leq \frac{23a_9}{87} \quad (B_{9,7} \text{ regime}).$$

Similarly, $a_2 \leq 23a_9/98$ in the $B_{9,8}$ regime and for $\sigma_9 = +1$.

α-regions: incompatibility.

- α_1: $20/9 > 23a_9/87 \iff a_9 < 580/69 \approx 8.41$.
- α_2: $20/9 > 23a_9/98 \iff a_9 < 1960/207 \approx 9.47$; throughout $[980/109, 9)$.
- α_3: same upper bound; incompatibility holds for $a_9 \in (9, 1960/207)$.

β-regions: residual right gap. When the constraints are compatible, the runners $B_{9,k}$ and $B_{2,1}$ together cover I_9 up to a residual right gap with left endpoint $g_L \geq 100/110$ in each case. Hence Theorem 5.5 applies. □

Theorem 6.5 (Positive-branch static core)**.** *In every completely blocking configuration with $\sigma_2 = +1$, $\mathrm{GF}(I_k) > 0$ for some slot I_k.*

Proof. In the α-regions of Lemma 6.4, the bounds from I_4 and I_9 are incompatible. In the β-regions, Theorem 5.5 (Universal Blocking) shows that the residual right gap either survives in I_9 or is absorbed only at the cost of opening another slot via PSM, Chain A, or Chain B. In all cases $\mathrm{GF}(I_k) > 0$ for some k. □

7. Phase-Uniform Closure in W_{ref}

7.1. Phase independence.

Lemma 7.1. *For every runner j and every cycle index m, $|B_{j,k}| = 2/(11a_j)$. Hence feasibility for covering a gap of fixed width depends only on a_j, not on phase.*

Proof. Immediate from the formula for $B_{j,k}$. □

Lemma 7.2. *The hypotheses and conclusions of PSM, Chain A, Chain B, the Negative-Branch Gap Lemma, and Theorem 6.5 depend only on the speed ranges $a_j \in (j-1, j+1)$ and on which runner covers the right gap of I_9. They are independent of the cycle index.*

Proof. In pacemaker-relative coordinates the slot boundaries I_s are fixed. In the m-th pacemaker cycle, runner j's relevant bad interval is B_{j,k_m} where $k_m = k_0 + m$. The residual fractional position $\{a_j t - k_m\}$ equals the cycle-0 residual up to an integer shift, so the slot-comparison inequalities are numerically identical in every cycle; the k-index is invisible to the endpoint arithmetic. Every inequality in Sections 5 and 6 is a comparison involving only a_j and the universal slot boundaries $\{(11s+i)/110\}$. The branch choice (PSM / Chain A / Chain B) is determined by which speed ranges hold and which runner covers the right gap of I_9; neither condition depends on the absolute phase. No step depends on a specific cycle index m. □

Remark 7.3. The phase-independence of the k-index is not merely notational. It means that the static-core analysis of Section 6 propagates unchanged to every complete pacemaker cycle inside W_{ref}, and there is no hidden dependency on which $B_{j,k}$ index is serving as the local primary branch in a given cycle.

7.2. **The phase-uniform static core.**

Theorem 7.4 (Phase-uniform static core)**.** *Let $C_m \subset W_{\text{ref}}$ be any complete pacemaker cycle. Then some slot $I_k^{(m)}$ has positive good fraction.*

Proof. Split on σ_2. If $\sigma_2 = +1$, Theorem 6.5 gives $\mathrm{GF}(I_k) > 0$ in cycle-relative coordinates, and phase-independence makes this phase-uniform. If $\sigma_2 = -1$, the Negative-Branch Gap Lemma gives a positive witness width in I_9 by a pure speed-range calculation, persistent in every cycle. Hence every complete pacemaker cycle inside W_{ref} contains a slot with positive good fraction. □

Corollary 7.5 (W_{ref} closure)**.** *In the commensurable large-ratio case $a_1 < 1$, there exists a witness in W_{ref}.*

Proof. Runner 1 is isolated throughout W_{ref} by definition. We need an integer m such that $C_m = [m/10, (m+1)/10] \subset W_{\text{ref}}$, i.e.

$$\frac{10}{11a_1} \le m \le \frac{100}{11a_1} - 1.$$

The length of this interval is $90/(11a_1) - 1 > 90/11 - 1 = 79/11 > 1$ because $a_1 < 1$. Hence it contains an integer m^*, so $C_{m^*} \subset W_{\text{ref}}$. By Theorem 7.4, some slot inside that cycle has positive good fraction for runners $2, \ldots, 10$. Runner 1 is safe everywhere on the cycle, giving a witness for all runners. □

8. The Irrational Case and the Full Proof

Lemma 8.1 (Commensurable reduction)**.** *If $q_1 < \cdots < q_{10}$ are positive commensurable reals, then after clearing denominators the problem reduces to the corresponding integer-speed configuration.*

Proof. Write $q_i = p_i \omega$ with integers p_i and $\omega > 0$. Then $\|q_i t\| = \|p_i(\omega t)\|$, so a witness for $(p_1, \ldots, p_{10})$ yields one for $(q_1, \ldots, q_{10})$. □

Logical note: Corollary 7.5, proved in §7, handles only the commensurable large-ratio case and is self-contained. There is no circularity.

Proof of the Irrational-Ratio Proposition. Let $\mathbf{x}(t) = (e_1 t, \ldots, e_{10} t) \bmod 1 \in \mathbb{T}^{10}$ and $\mathcal{S} = (1/11, 10/11)^{10}$. It suffices to show the orbit closure of $\mathbf{x}(t)$ meets $\mathcal{S}$.

Let $L = \{m \in \mathbb{Z}^{10} : m_1 e_1 + \cdots + m_{10} e_{10} = 0\}$ and let T_e be the Kronecker subtorus determined by L. Since some ratio e_i/e_j is irrational, $\dim_{\mathbb{Q}} \operatorname{span}_{\mathbb{Q}}\{e_1, \ldots, e_{10}\} \geq 2$.

Choose a $\mathbb{Q}$-basis $b_1, \ldots, b_d$ from $\{e_1, \ldots, e_{10}\}$, write $e_i = \sum_k r_{ik} b_k$ with $r_{ik} \in \mathbb{Q}$, pick a rational vector v close to $(b_1, \ldots, b_d)$, and define $q_i := \sum_k r_{ik} v_k$. By taking v sufficiently close the q_i are positive, distinct, ordered, and satisfy $q_{10}/q_1 \neq 10$.

If $q_{10}/q_1 < 10$, the interval $(1/(11q_1), 10/(11q_{10}))$ is nonempty; choose s^* there. Then $y := (\{q_1 s^*\}, \ldots, \{q_{10} s^*\}) \in \mathcal{S}$.

If $q_{10}/q_1 > 10$, Lemma 8.1 and Corollary 7.5 apply to $(q_1, \ldots, q_{10})$, producing an open witness set inside W_{ref}. Since $\mathcal{S}$ is open, any witness in that open set is strict; pick such a $y \in \mathcal{S}$.

In either case, the rational vector $v = (v_1, \ldots, v_d)$ was chosen within the rational hyperplane defined by the integer relations L of the original speed vector $(e_1, \ldots, e_{10})$: explicitly, any integer relation $\sum c_i e_i = 0$ (respectively $c_i e_i = c_j e_j$) continues to hold for the approximating speeds $q_i = \sum_k r_{ik} v_k$, because the coefficients $r_{ik} \in \mathbb{Q}$ are unchanged and v was perturbed within the same rational span. Consequently the orbit closure of $(q_1, \ldots, q_{10})$ under the torus flow lies in T_e, so $y \in T_e$. By Kronecker's theorem [5], the orbit $\{\mathbf{x}(t)\}$ is dense in T_e, so some $t > 0$ gives $\mathbf{x}(t) \in \mathcal{S}$. □

Proof of the Main Theorem. Every speed configuration falls into exactly one of three cases:

Case	Condition	Method
B	$e_{10}/e_1 \leq 10$	Bounded-Ratio Theorem
E	some $e_i/e_j \notin \mathbb{Q}$	Irrational-Ratio Proposition
R>	all $e_i/e_j \in \mathbb{Q}$ and $e_{10}/e_1 > 10$	Corollary 7.5

The three cases are mutually exclusive and exhaustive. Each is handled by the stated method. Hence the Main Theorem holds for all ten-tuples of positive speeds. □

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Appendix A. Technical Interval Arithmetic

This appendix collects the endpoint arithmetic that is routine but too repetitive for the main line. Sections A.1–A.8 are the eight slot-boundary ledgers used in Lemma 5.1, with A.5–A.8 written as standalone right-edge calculations. Section A.9 records the extracted 45/110 consequence used in Lemma 6.3. Section A.10 records the feasibility table behind

Universal Blocking. Section A.11 is the compact ledger for the Negative-Branch Gap Lemma.

A.1 Boundary ledger for $12/110$ **(slot** I_1 **left).** To straddle $12/110$, a branch $B_{j,k}$ must satisfy

$$\frac{11k+10}{11a_j} \le \frac{12}{110} \le \frac{11k+12}{11a_j}.$$

Inside the sign-range box $a_j \in (j-1, j+1)$, the only off-schedule candidates are:

branch k	speed interval	candidates
0	[25/3, 10)	runners 8, 9

- If runner 8 covers $12/110$, then $a_8 \ge 25/3$ and $B_{8,2}[0] = 32/(11a_8) \ge 32/99 > 34/110 = I_3[\text{left}]$, so the left edge of I_3 is missed.
- If runner 9 covers $12/110$, then either $a_9 < 86/9$ and $B_{9,3}[0] = 43/(11a_9) > 45/110$, or $a_9 \ge 86/9$ and $B_{9,3}[1] = 45/(11a_9) < 54/110$, so I_4 is not closed.

Thus $12/110$ cannot be repaired off schedule without opening another claimed slot.

A.2 Boundary ledger for $23/110$ **(slot** I_2 **left).**

branch k	speed interval	candidates
0	[100/23, 120/23]	runners 5, 6
1	[210/23, 10)	runner 9

- Runners 5 or 6 via $B_{j,0}$: $B_{j,0}[0] \ge 23/132 > 12/110$, so I_1 acquires a left gap that neither runner can close alone.
- Runner 9 via $B_{9,1}$: same split as in A.1 shows runner 9 fails to close I_4.

A.3 Boundary ledger for $34/110$ **(slot** I_3 **left).**

branch k	speed interval	candidates
0	[50/17, 3)	runner 2
0	[3, 60/17]	runner 4
1	[105/17, 115/17]	runners 6, 7
2	[160/17, 10)	runner 9

Each candidate either forces its pair-partner slot open or shifts runner 9's I_4 interval off a required boundary. In each case one obtains positive good fraction in a partner slot.

A.4 Boundary ledger for $45/110$ **(slot** I_4 **left).** This is the calculation used in Lemma 6.3.

branch k	speed interval	candidates
0	[20/9, 8/3]	runners 2, 3
1	[14/3, 46/9]	runners 4, 5, 6
2	[64/9, 68/9]	runners 7, 8
3	[86/9, 10]	runner 9

Runners 3–8 are each excluded: runner 3 forces $\mathrm{GF}(I_3) > 0$; runner 4 triggers Chain B opening I_7; runner 5 leaves I_1 open; runner 6 leaves I_8 open; runner 7 creates an inter-interval gap in I_7; runner 8 forces $\mathrm{GF}(I_3) > 0$. Runner 9 is excluded by the Lemma 6.3 left-gap analysis. Therefore runner 2 is the unique viable filler, giving $a_2 \ge 20/9$.

A.5 Boundary ledger for 65/110 (slot I_5 right).

branch k	speed interval	candidates
1	[42/13, 46/13]	runners 3, 4
2	[64/13, 68/13]	runners 4, 5, 6
3	[86/13, 90/13]	runners 6, 7
4	[108/13, 112/13]	runner 8

Every off-schedule repair of 65/110 forces positive good fraction in a partner slot by direct endpoint arithmetic (see the detailed exclusions in A.5 of the working notes).

A.6 Boundary ledger for 76/110 (slot I_6 right).

branch k	speed interval	candidates
0	[25/19, 30/19]	runner 2
1	[105/38, 3)	runner 2
1	[3, 115/38]	runner 4
2	[80/19, 85/19]	runners 4, 5
3	[215/38, 225/38]	runners 5, 6
4	[135/19, 140/19]	runner 7
5	[325/38, 335/38]	runner 9

Every off-schedule repair of 76/110 leaves a partner slot open.

A.7 Boundary ledger for 87/110 (slot I_7 right).

branch k	speed interval	candidates
0	[100/87, 40/29]	runner 2
1	[70/29, 230/87]	runners 2, 3
2	[320/87, 340/87]	runner 3
3	[430/87, 150/29]	runners 5, 6
4	[180/29, 560/87]	runner 6
5	[650/87, 670/87]	runner 8
6	[760/87, 260/29]	runners 8, 9

Runner 2 ($k = 0$): $B_{2,0}[\text{left}] \geq 29/44 > 65/110$, so runner 2 misses I_5. Runner 2 ($k = 1$): $B_{2,1}[\text{left}] \geq 1827/2530 > 56/110$, again missing I_5 left. Runners 3 ($k = 1, 2$): the next interval $B_{3,k+1}$ jumps above I_6 right, and runner 8 alone cannot cover I_6. Runners 5 and 6 ($k = 3$): the pair $(5, 6)$ loses I_8 right by incompatible bounds. Runner 6 ($k = 4$), runner 8 ($k = 5, 6$), runner 9 ($k = 6$): each forces a partner slot open by analogous endpoint arithmetic.

A.8 Boundary ledger for 98/110 (slot I_8 right).

branch k	speed interval	candidates
0	[50/49, 60/49]	runner 2
1	[15/7, 115/49]	runners 2, 3
2	[160/49, 170/49]	runners 3, 4
3	[215/49, 225/49]	runner 4
5	[325/49, 335/49]	runner 7
6	[380/49, 390/49]	runners 7, 8
7	[435/49, 445/49]	runner 9

Runner 2 ($k = 0$): $B_{2,0}[\text{left}] \geq 49/66 > 67/110$, so I_6 left is uncovered. Runners 2, 3 ($k = 1$): the previous interval lies below I_5 or I_6, opening those slots. Runner 4 ($k = 2, 3$): $B_{4,4}[\text{left}] > 98/110$ for $k = 3$, so runner 4 misses I_7. Runners 7, 8 ($k = 6$): start above I_6 right, leaving I_6 left uncovered. Runner 9 ($k = 7$): $445/49 < 86/9$, so Lemma 6.3 forces $a_2 \geq 20/9$, but covering $56/110$ of I_5 requires $a_2 \leq 15/7 < 20/9$; contradiction.

A.9 The full $45/110$ exclusion used in Lemma 6.3. If $a_9 < 86/9$ and I_4 is to be closed, runner 9 leaves the left edge $45/110$ uncovered. The A.4 exclusions show: runner 3 opens I_3; runner 4 opens I_7 via Chain B; runner 5 opens I_1; runner 6 opens I_8; runner 7 opens I_7 through its inter-interval gap; runner 8 opens I_3. Hence runner 2 is the unique viable filler and

$$\frac{10}{11a_2} \leq \frac{45}{110} \implies a_2 \geq \frac{20}{9}.$$

A.10 Feasibility table for Universal Blocking. For a right gap $[g_L, 109/110] \subset I_9$, the potentially feasible fillers are:

Runner	Sign	Feasible range	Conclusion
3	-1	$a_3 \leq 230/109$	PSM opens I_3
3	$+1$	$a_3 \in [32/(11g_L), 340/109]$	Chain A opens I_8
4	-1	$a_4 \leq 340/109$	PSM opens I_2
4	$+1$	$a_4 \in [43/(11g_L), 450/109]$	Chain B opens I_7
5	-1	$a_5 \leq 450/109$	PSM opens I_1
5	$+1$	no branch survives	no cover
6	± 1	no branch survives	no cover
7, 8	± 1	no branch survives	no cover

More explicitly, for runners 5–8 with the positive sign (or any sign for runners 6–8), the bounds

$$\frac{11k + 10}{10} > \frac{10(11k + 12)}{109} \quad \text{for all } k \geq 2$$

rule out every feasible branch.

A.11 Compact ledger for the Negative-Branch Gap Lemma. When $\sigma_2 = -1$, the Runner-2 Elimination Lemma forces $a_9 = 86/9$ with $B_{9,8}(86/9) = [441/473, 450/473]$. The cascade-forced outer speeds by subcase:

- $\sigma_3 = +1$: $a_3 = 50/17$, $a_8 = 140/19$.
- $\sigma_3 = -1$: $a_3 = 115/38$.
- $\sigma_4 = +1$: $a_4 = 100/23$, $a_7 = 560/87$.
- $\sigma_4 = -1$: $a_4 = 340/87$.

The four subcases are exactly the table in the Negative-Branch Gap Lemma, and the minimal uncovered width is

$$\frac{1216}{1265} - \frac{450}{473} = \frac{538}{54395}.$$

Appendix B. Computational Corroboration

The computations described here are *not* part of the proof; the theorem is established analytically in Sections 1–8. They provide independent, large-scale validation that the proof's structural mechanisms behave exactly as claimed, with no hidden exceptional regimes detected.

Exact symbolic verification.

- Exact rational checks on the canonical integer family $\{1, \ldots, 9, p_{10}\}$ with $p_{10} \in \{89, \ldots, 200\}$ find witnesses in W_{ref} in all 112 cases.
- All 34 explicit rational inequalities in the proof—including the endpoint constants
$$\frac{83}{2695}, \quad \frac{17}{430}, \quad \frac{1}{50}, \quad \frac{1533}{18490}, \quad \frac{538}{54395}$$
and the Chain A, Chain B, and negative-branch bounds—were verified with exact arithmetic in Python (`fractions.Fraction`), giving zero floating-point error. All 34 checks pass.

Large-scale boundary-set verification. Two independent witness-search campaigns were conducted on an Apple M4 MacBook Pro (10-core CPU, Python 3.14). The event-complete boundary-set method of [6] was used: rather than sampling a time-grid, the search evaluates only at the finite set of times where some runner's fractional distance to the origin crosses the 1/11 threshold, guaranteeing that no witness in W_{ref} can be missed between evaluation points. This approach, developed in the context of the $\phi(n)$ law for arithmetic progressions [6], reduced per-configuration cost by a factor of approximately 10^3 compared with naive grid search and made the following scale computationally feasible with no cloud resources.

Campaign	Seed	Configurations	Failures	Boundaries evaluated	Runtime
`run_1e8_mech`	42	10^8	**0**	6.1×10^{10}	65 min
`run_1e9_mech`	1337	10^9	**0**	6.1×10^{11}	10.7 hr
Combined	—	1.1×10^9	**0**	6.7×10^{11}	—

Each configuration was drawn randomly from the commensurable large-ratio regime ($a_1 \in (0, 1)$, a_j uniformly in its sign-range interval for $j = 2, \ldots, 9$, $a_{10} = 10$) and classified into one of the eight analytic regions of the proof ($\sigma_2 = \pm 1$; sub-regions $\alpha_1, \alpha_3, \beta_1, \beta_2, \beta_3, \beta_4$). Every region was covered in both campaigns and no exceptional behaviour was observed in any region.

With zero failures across 1.1×10^9 event-complete trials, the 99.9% confidence upper bound on the true failure rate is 6.9×10^{-9}. For comparison, the probability of a hardware bit-flip error during the run exceeds this bound, making undetected software error the dominant residual uncertainty.

All scripts, raw results, and reproduction instructions are available at:
`https://github.com/trentjp-gecta9/lrc-n11-computational-verification`

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